

Q1

$$a = 4, b = 3/8, c = -2$$

Q2

$$x = [4 \pm \sqrt{16 + 12(2+\sqrt{5})(2-\sqrt{5})}] / [2(2+\sqrt{5})]. \text{ Since } (2+\sqrt{5})(2-\sqrt{5}) = 4-5 = -1, \text{ this simplifies to } x = (4 \pm \sqrt{16-12}) / (2(2+\sqrt{5})) = (4 \pm 2) / (2(2+\sqrt{5}))$$

Q3a

$$\text{leading coefficient} = +3, \text{ one factor} = (x+2), \text{ remaining factors} = (x-1)(x-4)$$

Q3b

$$\text{Case 1: } 5x-2 \geq 4x+1 \text{ gives } x \geq 3 \text{ (critical value 3). Case 2: } 5x-2 \leq -4x-1 \text{ gives } 9x \leq 1 \text{ so } x \leq 1/9 \text{ (critical value } 1/9). \text{ Combined solution: } 1/9 \leq x \leq 3$$

Q4a

$$r = 12 \text{ and } \frac{1}{2}r^2 = 57.6. \text{ Dividing the second by the first: } \frac{1}{2}r = 57.6/12 = 4.8, \text{ so } r = 9.6. \\ \text{Then } = 12/9.6 = 1.25$$

Q4b

$$AC = r \tan = 9.6 \times \tan(1.25) = 28.89. \text{ Shaded area} = (\frac{1}{2} \times 28.89 \times 9.6) - 57.6 = 138.67 - 57.6 = 81.1$$